Step 1: Is the System an "AI System"?

How to:

The first step for understanding the potential obligations under the AI Act is to find out whether the System under analysis meets the definition of "AI System" in the meaning of the AI Act.

If it is indeed an AI System, continue with step 2.

What you need to know about the System before getting started:

- Intended purpose
- Approach for implementation
- System Output

What do you think?

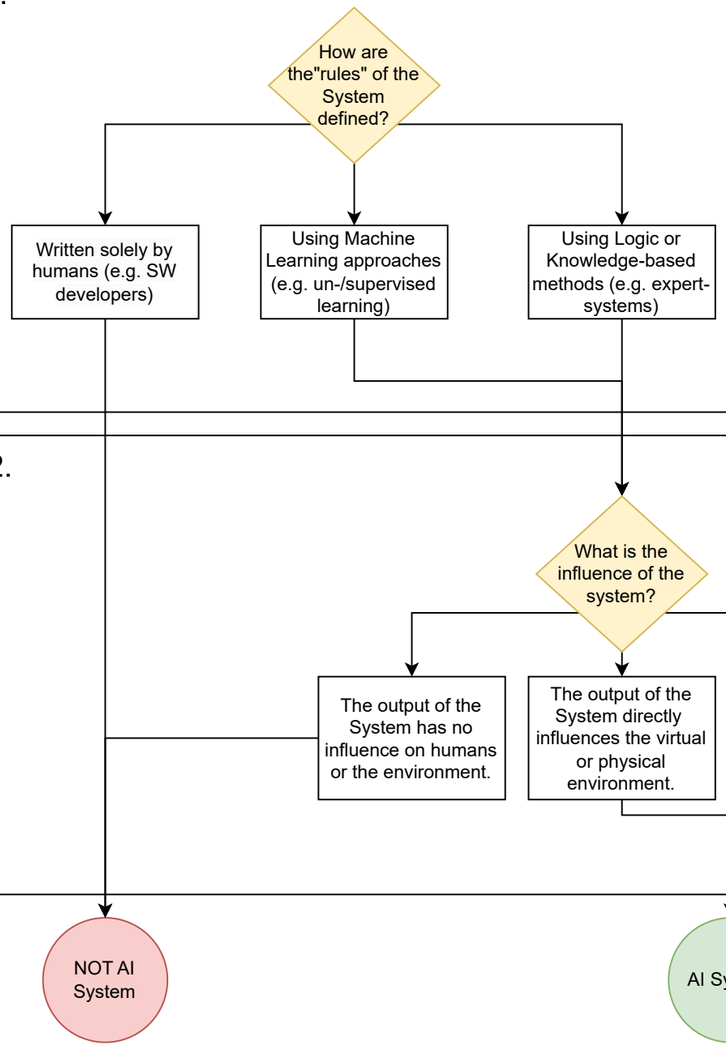
We omitted two aspects from the AI Definition in this decision tree, because they do not seem to matter in for the decision.

- Elements of Autonomy: Any System that is solely based on rules written by a human has, by implication, no elements of autonomy. In contrast, Systems using Machine Learning approaches or Logic or Knowledge based methods operate, by implication, with some elements of autonomy.
- Output: The outputs listed in the AI Act seem to illustrateAI Systems, but they are not an specific discriminator that goes beyond decision 1.

Do you agree or can you think of counter-examples? Let us know. Click here or scann this QR Code.

QR Code to the feebdack questionnaire

1.



Guidance

Origin of the "rules"

A distinguishing feature of AI Systems is that the "rules of how they are functioning" are not solely written by humans, as it is the case with software programs, but that the rules are inferred from data, either through machine learning or logic and knowledge based approaches.

In which of those three approaches does your system fall?

- **software** systems and programming approaches whose rules are written by humans to automatically execute operations. Software systems solely written by humans are not AI.
- **machine learning programs** that are capable of learning and inferring from data i.e: supervised, unsupervised and reinforcement learning. They use a variety of methods such as neural network, logistic regression, Bayesian estimation, and search / optimization methods (see also Recital 6a).
- **logic and knowledge based** approach utilize encoded rules, ontologies or knowledge graphs by humans to infer new information through sorting, searching, matching or chaining. i.e: Logic programming, symbolic reasoning, expert systems, and search and optimization methods (see also Recital 6b).

References & Examples

References: AI Act Article 3 (1), Recital 6, 6a-b

Simple Software: "shopping cart in an online store, how to proceed from the item selection to the checkout"

Machine Learning: AI trained on annotated contract archives to extract contractual metadata, attributes, and clauses on a large scale. With the help of OCR (Optical Character Recognition) technology contracts can be converted into analyzable digital assets by unstructured text blocks

Logic/knowledge based: Expert system that uses a set of rules to diagnose a disease based on patient symptoms. AI system which recognizes new types of fraud through detection of anomalies. Exploratory statistical analysis in the data helps to define the rules which will flag anomalies for unforeseen system behavior. Path finding and route optimization for road traffic.

2.

Area of Influence

Lastly, the AI Act focusses on Systems whose output is intended to influence the environment, both physical and virtual, as well as Systems that influence the actions or decisions of humans. Recital 6 states "The concept of the autonomy of an AI system relates to the degree to which such a system functions without human involvement.". Systems that influence neither of those, are not considered an AI System in the meaning of the AI Act.

What does the system under analysis influence?

- **Environment: physical or virtual**
- **Humans: action or decision**
- **Neither of the above**

References: AI Act Article 3 (1), Recital 6

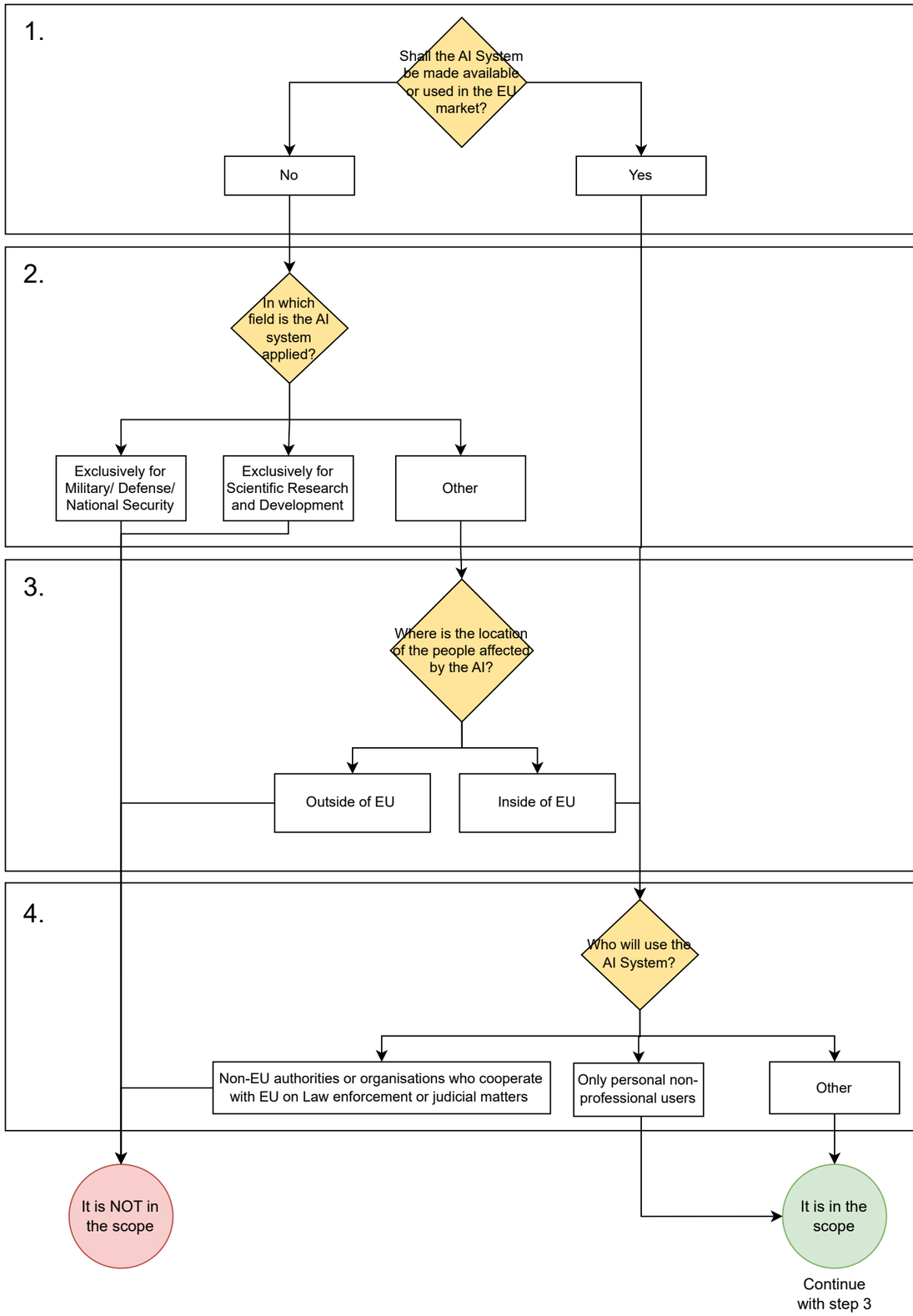
Examples:

- AI System that influence its **environment:** Quality Control in a factory, where the AI System actuates a machine to separate "OK" from "Not OK" products.
- AI System that directly influences on **humans** behaviour: A recommendation system that promts the user to go to certain places e.g. when visting a city.
- AI System that influences **neither the environment nor humans**: A generative AI System that is used for creative / artistic purposes, e.g. art installments.

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Step 2: Is the AI System in scope of the EU AI Act?

The second step foconcerns the question whether the AI system is in the scope of the EU AI Act or not. If it is inside the scope, continue with Step 3. As an input, you need to know the location of the user and the affected persons of the AI System, as well as the intended purpose.

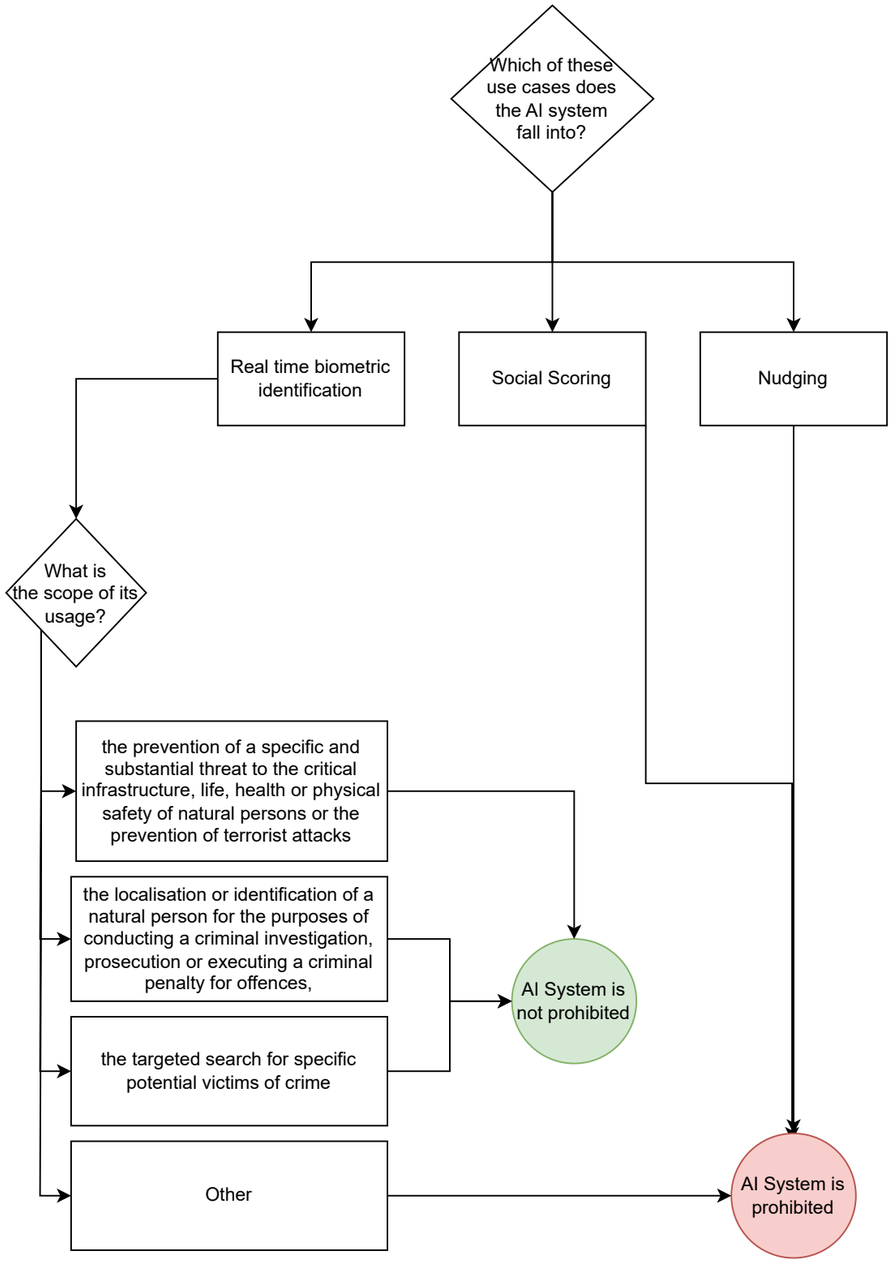


Guidance	References & Examples
Location of Usage The AI Act applies to • providers of AI Systems who make such systems available on the EU market or put them into service in the EU. • users of AI Systems who are physically present in the EU	See Article 2 for the entire Scope. Useful definitions in Article 3: • '*placing on the market*' means the first making available of an AI system on the Union market; • '*making available on the market*' means any supply of an AI system for distribution or use on the Union market in the course of a commercial activity, whether in return for payment or free of charge; • '*putting into service*' means the supply of an AI system for first use directly to the user or for own use in the Union for its intended purpose
Area of application Some areas of application are excempt from the scope of the AI Act, which is especially the case, when they are not being made available on the EU market. Specificially, off-scope are AI Systems with exclusive usage in: • military, defence or national security (Article 2. (3)) • scientific research and development (Article 2. (6)) • research and development activity regarding AI systems (Article 2. (7))	See Article 2 for the entire Scope. Recital 12a and 12b explain the exemptions for military and science. Useful definitions in Article 3: • 'intended purpose' means the use for which an AI system is intended by the provider, including the specific context and conditions of use, as specified in the information supplied by the provider in the instructions for use, promotional or sales materials and statements, as well as in the technical documentation;
Location of affected persons The protection of natural persons in the EU is a central objective of the AI Act. Hence, AI systems that are operated outside the EU may also fall into the scope of the AI Act, if the output of the AI System affects natural persons inside the EU According to Article 2, the AI Act applies to "(...) providers and users of AI systems who are physically present or established in a third country, where the output produced by the system is used in the Union;".	Recital 11: "(...) to ensure an effective protection of natural persons located in the Union, this Regulation should also apply to providers and users of AI systems that are established in a third country, to the extent the output produced by those systems is used in the Union." See also Article 3 for the definitions of (9) placing on the market (10) making available on the market (11) putting into service Example: A medical laboratory in the UK operates a rare AI System for diagnistic purposes and a EU-based hospital sends a tissue sample of a local patient to the lab.
User Group The question of "who uses the AI System?" has an effect on the applicability of the AI Act: • Non-EU authorities or other organisations who collaborate with the EU or members states in the context of Law Enforcement or judicial matters • Non-professional users are in the scope, but the obligations are limited to Article 52.	Recital 11: "(...) this Regulation should not apply to public authorities of a third country and international organisations when acting in the framework of international agreements concluded at national or European level for law enforcement and judicial cooperation with the Union or with its Member States." See also Article 3 for the definitions of (39) Law Enforcement (40) Law Enforcement authority Example: The Use of AI in a collaboration between the EU and INTERPOL.

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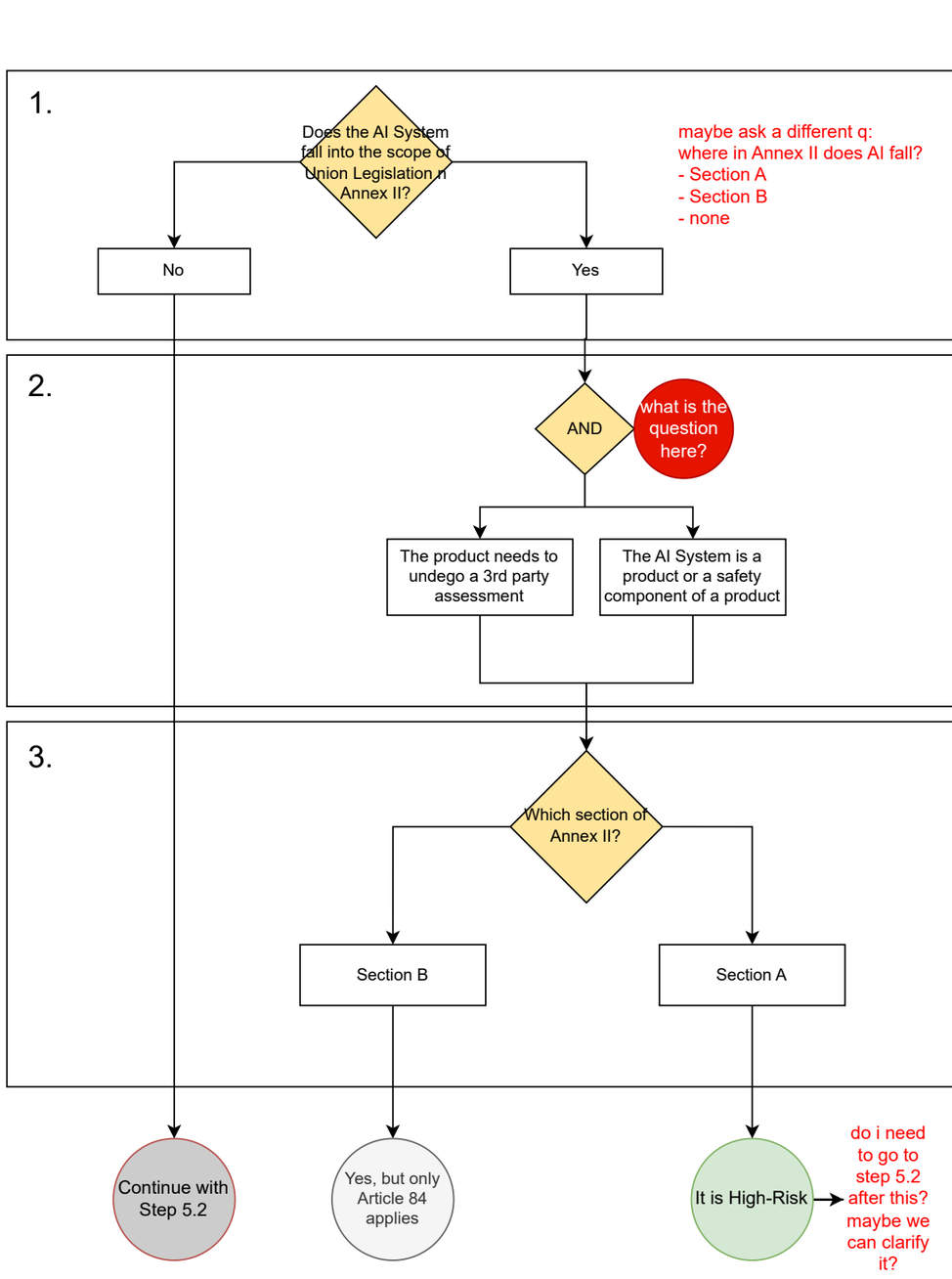
Step 3: Is the AI System prohibited?

The third step is for assessing whether the AI System falls under any prohibited categories depending on the use case



Step 5.1: Is the AI System a High-Risk System? (Annex II)

The fith step is helps you to assess whether the AI System is High-Risk with a focus on existing Union Legislation. In any case, check also whether Step 5.2 applies, which is about High-Risk in standalone applications.



Guidance	References & Examples
Existing Union Legislation what if we combine this column with the other one? for examples we can specify the guidance by mentioning what directive refers to: i.e machine, toys, etc Work in Progress	
Qualification as High-Risk tbd	
Exemption of Annex II, Section B tbd	

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Step 5.2: Is the AI System a High-Risk System? (Annex III)

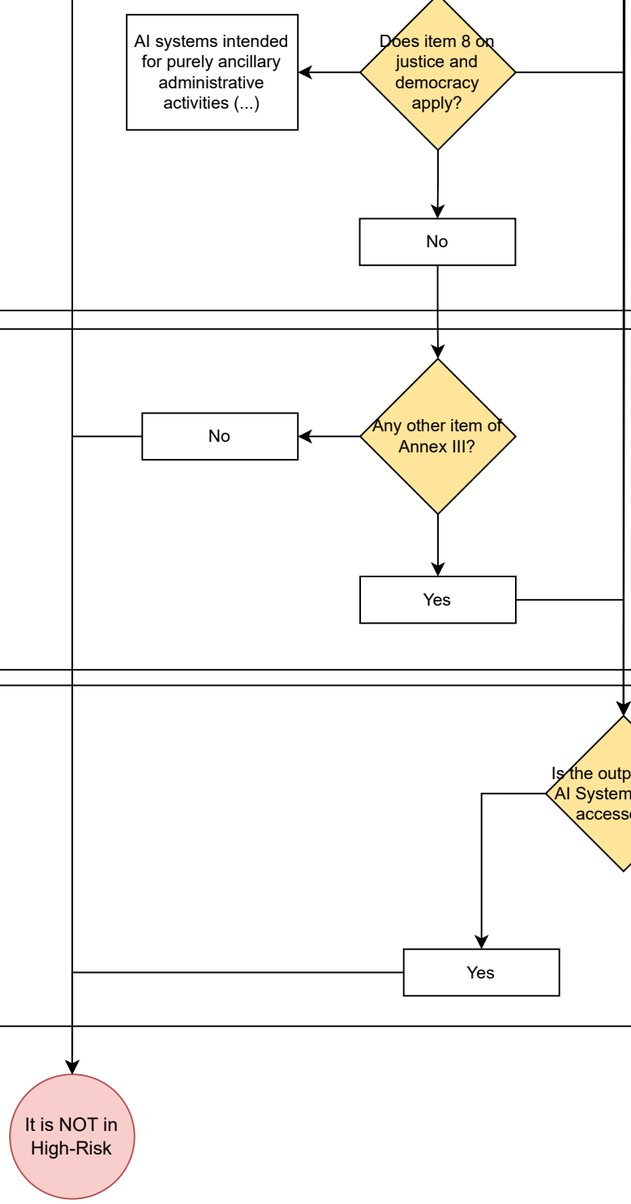
The fith step is helps you to assess whether the AI System is High-Risk with a focus on Standalone AI Systems.

where in the Annex III are these items? i cannot find them

Guidance	References & Examples
Standalone AI Systems maybe attach here Annex II with subcategories on the back?	
Exemptions for critical infrastructure tbd	
Exemptions for essential services tbd	
Exemptions for Law Enforcement	

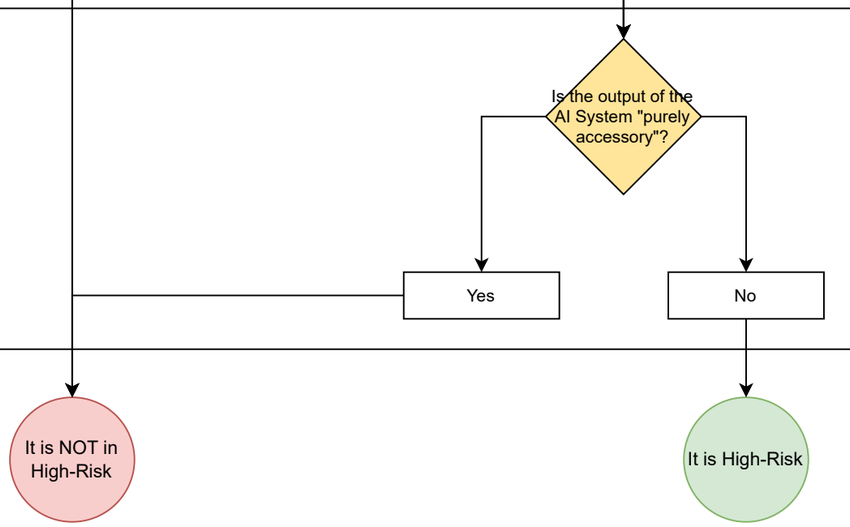
Work in Progress

5.



Exemptions for Justice and Democracy

Other Standalone AI Systems



Horizontal layer

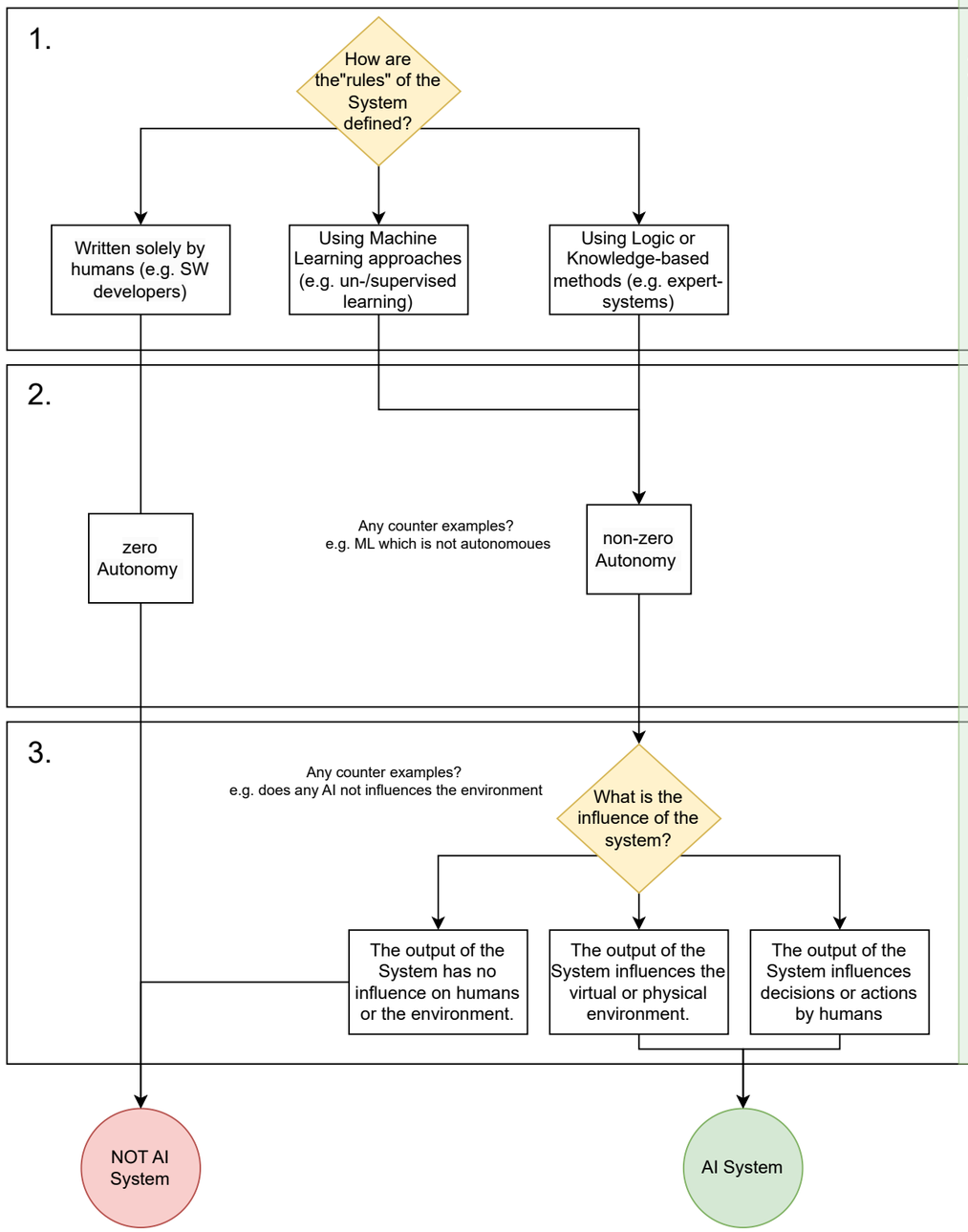
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Step 1: Is the System an "AI System"?

The first step for understanding the potential obligations under the AI Act is to find out whether the System under analysis meets the definition of "AI System" in the meaning of the AI Act.

What you need to know about the System before getting started:

- Intended purpose
- Approach for implementation
- System Output
- Level of Autonomy
- User
- Context of usage



Guidance

Origin of the "rules"

A distinguishing feature of AI Systems is that the "rules of how they are functioning" are not solely written by humans, as it is the case with software programs, but that the rules are inferred from data, either through machine learning or logic and knowledge based approaches.

In which of those three approaches does your system fall?

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Level of Autonomy

AI Systems are further differentiated by their ability to act autonomously, hence the AI Act considers something an AI System if it "operates with elements of autonomy". Consequently, a System with zero-autonomy is not considered an AI System.

Does the System under analysis operate with elements of Autonomy?

- **Yes (=non-zero autonomy)**
- **No (=zero-autonomy)**

To find out, refer to the general levels of autonomy on right or a sector-specific, if applicable. "Autonomy matters also when the System does some form of ranking, filtering, selection or recommendation"

Area of Influence

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References & Examples

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References: AI Act Article 3 (1), Recital 6a, 6b

ISO/IEC 22989:2022(en) refers to Autonomy as the "characteristic of a system that is capable of modifying its intended domain of use or goal without external intervention, control or oversight"

Autonomy Levels:

1. b
2. b
3. b
4. b
5. b
6. b

Sector-specific autonomy levels:
Mobile Networks, Automotive, Medical Devices, Education

References: AI Act Article 3 (1), Recital 6

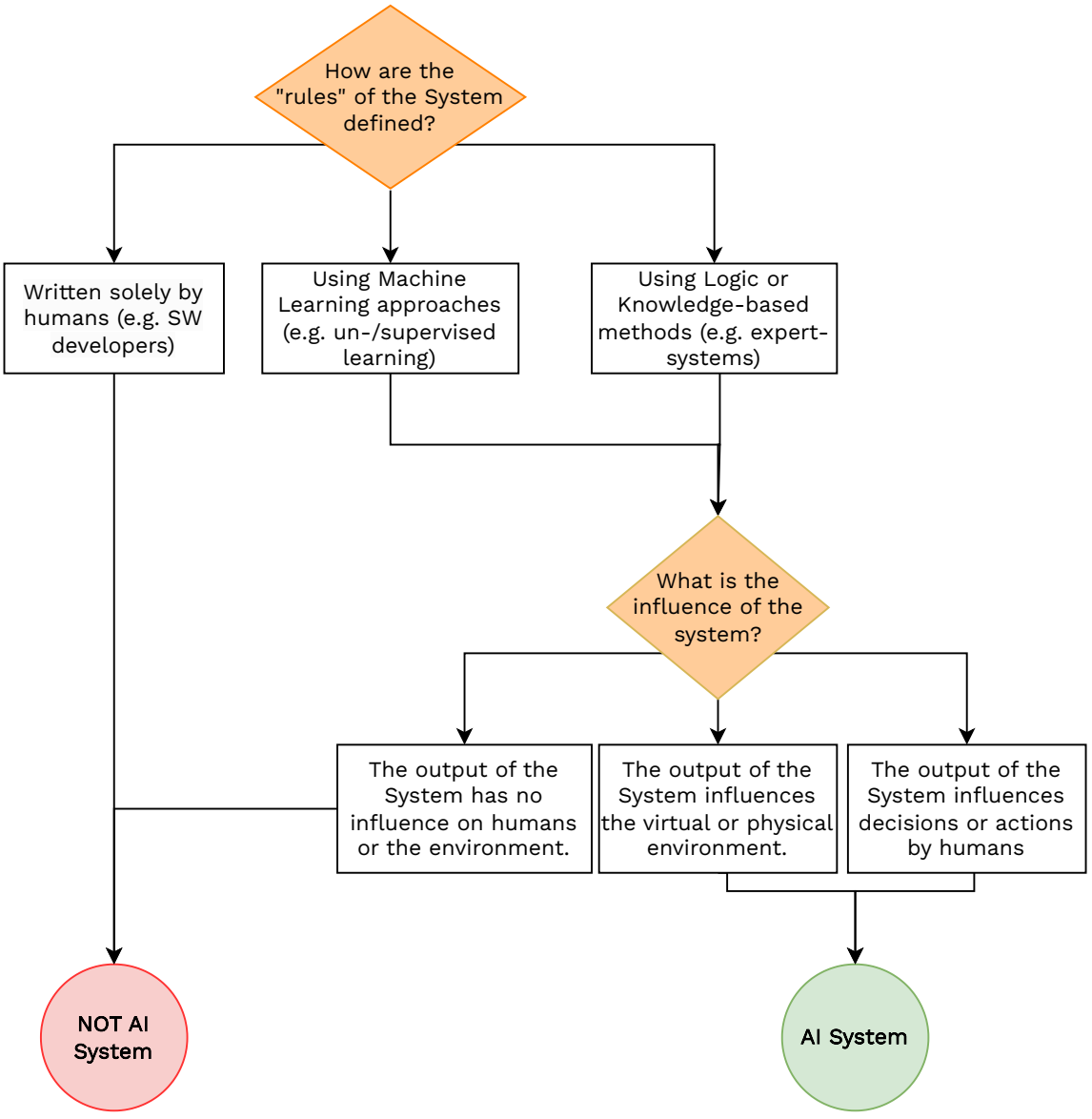
AI Systems that influence its **environment** include, for instance XXX

An influence on **humans** given in cases such as XXX

Areas of influence that are **not covered** by the AI Act include XXX.

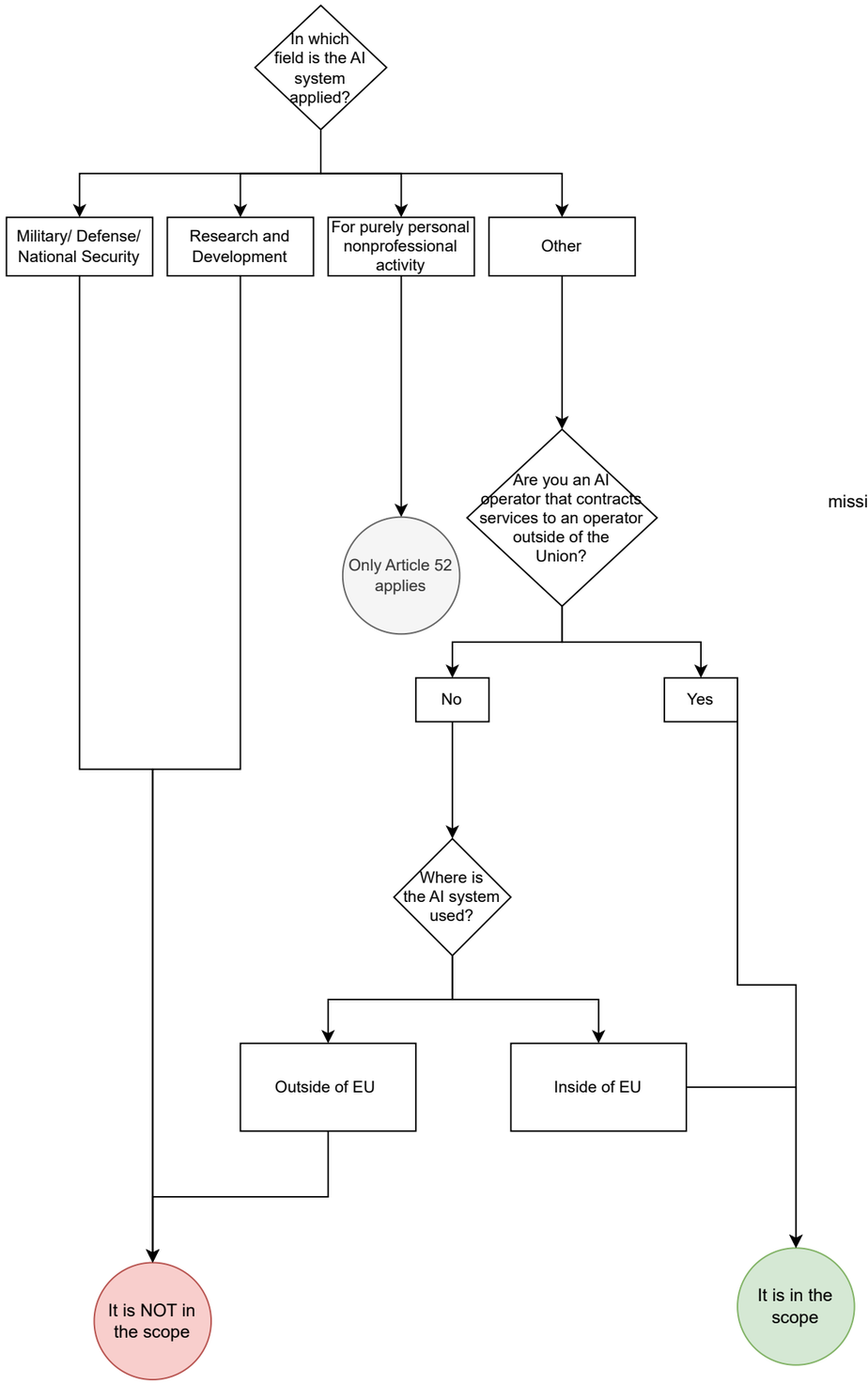
*__Automatic__ means performing a task according to pre-defined rules.
*__Autonomous__ means learning from the surroundings and deciding independently how to act.

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How are the "rules" of the System defined?
Written solely by humans (e.g. SW developers)
Using Machine Learning approaches (e.g. un-/supervised learning)
Using Logic or Knowledge-based methods (e.g. expert-systems)
What is the influence of the system?
The output of the System has no influence on humans or the environment.
The output of the System influences the virtual or physical environment.
The output of the System influences decisions or actions by humans
NOT AI System
AI System

Step 2: Is the AI System in scope of the EU AI Act?

The second step for understanding if the AI system is in the scope of the EU AI Act depending on the location, the area of application or the type of entity using the AI system



TBD:
- ART 2(2) - loophole
- ART 2 (4)

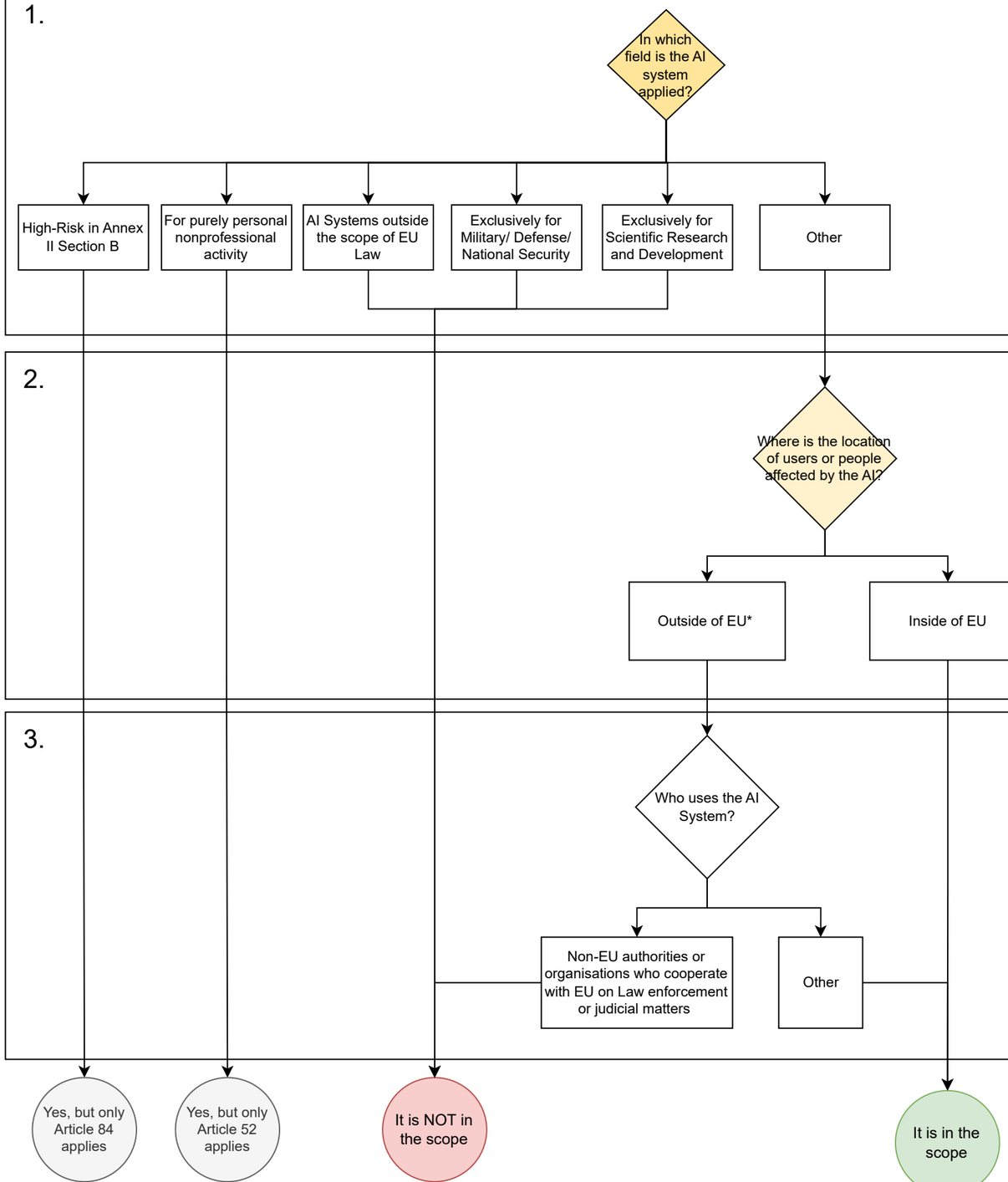
missing: when ppl in the EU Are affected by an output form an non_EU AI System

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Step 2: Is the AI System in scope of the EU AI Act?

The second step for understanding if the AI system is in the scope of the EU AI Act depending on the location, the area of application or the type of entity using the AI system

INITIATIVE FOR APPLIED ARTIFICIAL INTELLIGENCE



Guidance	References & Examples
Area of application High-Risk in Annex II, Section B: See step 4 Non-professional usage: See Article 2.8 and Recital 58	

* including EU public authorities outside the EU and international organisations who use the AI System in the context of an international agreement with one/more EU members states or the EU

Continue with step 3

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